NAME: ______________________ PERIOD: ________ DATE: ____________

Two Test Plans for Cure Time

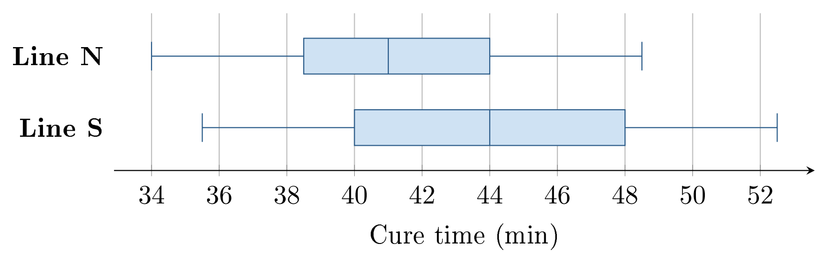
AP Statistics · Unit 4 · Topic 4.9 · B: 2027-01-28 · E: 2027-03-17

[STAR] TODAY'S PROBLEM

Suppose line N uses a new formula for sealant, a material that closes gaps. Line S uses the standard formula. A team takes independent simple random samples of 18 of 600 recent N batches and 18 of 540 recent S batches. Mean cure times are 41.2 and 44.0 minutes; the boxplots are shown. Before sampling, the team asked, "Do the two populations differ in mean cure time?" Formula was not randomly assigned to lines.

Rhea: "Keep the original question and use population means. A difference between lines may have other causes."

Sol: "Since $41.2 < 44.0$, use a lower tail with sample means. A significant result proves the formula caused faster curing."



FIRST TAKE — before the video (5 min)

Did the team originally ask whether N was faster or whether the lines differed in either direction? Explain in one sentence.

[BOOK] HYPOTHESES FOR TWO MEANS (keep on the page)

For two independent groups with a quantitative response, use a two-sample t test for $\mu_1 - \mu_2$. The null states no population difference: $H_0 : \mu_1 - \mu_2 = 0$. Choose $<$ or $>$ only when the research question specified that direction before seeing the data; otherwise use $\neq$. Hypotheses use population parameters, never observed sample means. Check independent random samples or random assignment, both 10 percent conditions when sampling without replacement, and both sample shapes when either $n < 30$. Random sampling supports population scope; random assignment is needed to isolate a causal treatment effect.

Finish after the video:

DOK 1 **(a)** Define μ_N and μ_S in context.

DOK 2 **(b)** State H_0 and H_a in N-minus-S order to match the question asked before sampling.

DOK 3 **(c)** In 3–4 sentences, judge the two plans. Explain why the original question sets the tail, why hypotheses use population means, and why these two lines cannot isolate the formula as the cause.

Frame: The original question asks ____________________, so the alternative should ____________________.

Frame: The hypotheses describe ____________________. A causal claim is limited because ____________________.

EXIT — turn this in

One thing the video changed about my first take: ____